

Doc. 315 – The Form of K_{frak} : Additive or Multiplicative?
With Euler's Musical Spiral (5/7-Limit) as Control Case

Johann Pascher

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.1 What this is about

The fractal correction factor appears throughout the corpus in the additive form

$$K_{\text{frak}}^{\text{add}} = 1 - 100\xi = \frac{74}{75} = 0.98\bar{6},$$

while a multiplicative–cumulative form would be conceivable:

$$K_{\text{frak}}^{\text{mul}} = (1 - \xi)^{100} = 0.9867543\dots$$

The relative distance between the two forms is $8.88 \cdot 10^{-5}$. The corpus confirms the *value* $74/75$ several times independently (A040, A130, A270); whether this also discriminates the *form* – additive against multiplicative – had remained unexamined. This document supplies the examination and answers two questions: Which corpus location can resolve the two forms? And – as a structural question – *why* should one form be the right one?

Euler's musical spiral serves as the control case. It is the classic example of a cycle that *never* closes and thereby provides the counter-probe to the ξ cycle, which closes *exactly* after 75 revolutions. The comparison of the two cases supplies, in the end, the conceptual axis on which the form question hangs: rolled-up versus unrolled description.

Preliminary clarification (avoiding a category error): The value $74/75$ is **[K]** by the A040 derivation; the question of this document concerns exclusively whether independent witnesses can *measure* the additive against the multiplicative form – i.e. the status of the form as a measured rather than merely derived statement.

.2 Control case I: Euler's spiral never closes

The impossibility argument

[Q] Euler's Tonnetz (*Tentamen novae theoriae musicae*, 1739) spans the intervals of just intonation as a lattice of fifths ($3/2$) and major thirds ($5/4$); the octave (2) is divided out. The question "after how many repetitions does the spiral close?" has an exact answer: *never*. A closure would require

$$2^x \cdot 3^y \cdot 5^z = 1 \quad \text{with integers } (x, y, z) \neq (0, 0, 0),$$

contradicting the uniqueness of prime factorisation. Equivalently: $\log 2$, $\log 3$, $\log 5$ are linearly independent over $\mathbb{Q}$. The orbit lies dense on the octave circle but never meets its starting point.

Near-closures in the 5-limit

[Q] What exists instead are near-closures whose deficit is in each case a named comma:

Path on the Tonnetz	Residual (comma)	Ratio	Cents
8 fifths + 1 maj. third vs. 5 octaves	schisma	$32805/32768$	≈ 1.95
4 fifths vs. third + 2 octaves	syntonic comma	$81/80$	≈ 21.51
12 fifths vs. 7 octaves	Pythagorean comma	$531441/524288$	≈ 23.46
3 maj. thirds vs. 1 octave	lesser diesis	$128/125$	≈ 41.06

The practically best closure in the 5-limit is the schisma (≈ 2 cents, below the hearing threshold); the best-known is the circle of fifths after 12 fifths with the Pythagorean comma as residual.

Near-closures in the 7-limit (script result)

[K] Adding the natural seventh $7/4$ as a third axis (lattice $\mathbb{Z}^3$ over 3, 5, 7), the same impossibility argument applies. The systematic search (euler_spirale_7limit.py, exact fraction arithmetic, search space $|b| \leq 12$ fifths, $|c| \leq 6$ thirds, $|d| \leq 4$ sevenths; an assertion confirms along the way that no exact closure ever occurs) yields near-closures an order of magnitude better than anything in the 5-limit:

Cents	Comma	Path $3^b 5^c 7^d$	Name
0.325	$250047/250000$	$b = -6, c = 6, d = -3$	landscape comma
0.396	$4375/4374$	$b = -7, c = 4, d = 1$	ragisma
0.721	$2401/2400$	$b = -1, c = -2, d = 4$	breedsma
1.954	$32805/32768$	$b = -8, c = -1, d = 0$	schisma (5-limit, for comparison)
5.758	$5120/5103$	$b = -6, c = 1, d = -1$	hemifamily
7.712	$225/224$	$b = -2, c = -2, d = 1$	septimal kleisma

The ragisma is the simplest sub-cent closure (7 fifths down, 4 thirds and 1 seventh up miss the starting point by 0.4 cents); the coarser 7-limit counterpart of the syntonic comma is Archytas' comma $64/63$ (≈ 27 cents). More prime axes make the lattice denser – the spiral approaches itself faster, but never closes in principle.

How the spiral is closed after all: temperament

[B] A closure only arises when commas are *tempered out*, i.e. declared equal to 1. Twelve-tone equal temperament sets the fifth by decree to exactly $7/12$ of the octave – it *rationalises* the step size, and only thereby does the circle of fifths close. The closed circle is not a property of the pure intervals but an artefact of rationalisation. This sentence becomes load-bearing in Section .6.

.3 Control case II: the xi cycle closes exactly

[B] The ξ cycle stands on the other side of the same dichotomy. The step size is

$$100\xi = \frac{400}{30000} = \frac{1}{75} \quad \text{– exactly rational,}$$

because $\xi = 4/30000$ is a rational stipulation (A020). A step of $1/75$ of a cycle closes after exactly 75 repetitions with zero residual; $K_{\text{frak}} = 1 - 1/75 = 74/75$ is the closed path minus one segment (A040, A050). This is the general torus dichotomy: rational winding number $\Rightarrow$ closed orbit; irrational $\Rightarrow$ dense, never-closing orbit.

	Musical spiral (5/7-limit)	ξ cycle (FFGFT)
Step size	$\log_2(3/2), \log_2(5/4), \dots$ irrational	$1/75$ rational
Reason	primes mult. independent	ξ stipulated rational
Behaviour	dense, never meets itself	closes exactly after 75
"Near-closure"	commas (schisma, ragisma, ...)	none needed
Verification	numerical only (cents)	exact fraction arithmetic

The closure after 75 is thus no numerical accident but a direct consequence of the rationality of ξ – the same rationalisation that music must impose after the fact through temperament is stipulated in the framework from the outset.

.4 The form question

[B] The two forms are identical to first order; their entire difference is the second binomial term:

$$(1 - \xi)^{100} = 1 - 100\xi + \binom{100}{2}\xi^2 - \dots, \quad K^{\text{mul}} - K^{\text{add}} = 4950\xi^2 - O(\xi^3) = 8.76 \cdot 10^{-5}.$$

Numerically verified: the binomial term $4950\xi^2 = 8.8 \cdot 10^{-5}$ covers 99.6 % of the full distance; the remainder is higher-order. A corpus location can decide the form only if its own uncertainty is smaller than this distance – for power-law locations: smaller than the correspondingly amplified distance.

.5 Three witnesses

Witness A: the α point (A130)

[K] A130 leads to the scale E_0 by two closely related routes: route 1 as the bare geometric mean $E_0 = \sqrt{m_e m_\mu} = 7.348$ MeV, route 2 as the same mean with the fractal correction, $E_0 = \sqrt{m_e m_\mu / K_{\text{frak}}} = 7.397$ MeV. They differ only by K_{frak} . The measured α fixes this factor without using it:

$$K_{\text{mess}} = \xi m_e m_\mu \alpha^{-1} = 0.98650.$$

[K] Finding: the residual against the additive form is $1.7 \cdot 10^{-4}$, against the multiplicative $2.6 \cdot 10^{-4}$. Both lie far above the form distance $8.9 \cdot 10^{-5}$; the location has a residual of its own that is larger than the distance between the forms. Witness A confirms the *value* 74/75 to $1.7 \cdot 10^{-4}$ but makes *no form statement*.

[K] Register status (R72): The register books the two E_0 values as *the same value bare and fractally corrected* (7.348 bare, 7.398 α -anchored; ratio $1/\sqrt{K_{\text{frak}}}$ by the A040 rule). A determination of the *exponent* follows from the ratio: $n = 101.2$, alongside $n = 100 \pm 2$ (R67) and $n = 100.27$ (sector ladder A270). In this n language the form question reads: additive $K = 1 - n\xi$ versus multiplicative $(1 - \xi)^n$ differ by $\Delta n_{\text{eff}} = n(n-1)\xi/2 \approx 0.66$ – none of the three n determinations currently reaches this sharpness.

Witness B: the high-power location (A270)

[K] A270 carries the identity $K_{\text{frak}}^{-36} \approx 16/\pi^2$ with documented agreement 0.010 %. The exponent 36 amplifies the form distance to $36 \cdot 8.9 \cdot 10^{-5} = 0.32$ % – far above this tolerance. The location is therefore resolution-capable:

Expression	Value	rel. dev. from $16/\pi^2 = 1.6211389$
$(74/75)^{-36}$ (additive)	1.6213007	+ $1.0 \cdot 10^{-4}$
$((1 - \xi)^{100})^{-36}$ (multipl.)	1.6161261	- $3.1 \cdot 10^{-3}$

Verdict: 31:1 in favour of additive – conditional on the assumption that $16/\pi^2$ is the correct reference. In A270 this is carried as a coincidence without forward derivation (P35), but Monte-Carlo secured.

[K] Upgrade of the reference through Doc. 314: The constant is identifiable corpus-natively. $\pi^2/16$ is the *packing density of the D4 lattice* (Doc. 314: "kissing number 24 against 8, packing density $\pi^2/16$ against $\pi^2/32$ ") – hence

$$\frac{16}{\pi^2} = \frac{1}{\Delta(\text{D4})}.$$

The A270 identity thus reads $K_{\text{frak}}^{-36} = 1/\Delta(\text{D4})$: the reciprocal packing density of precisely the lattice whose structure Doc. 314 verifies in Hilbert space. The reference is therefore no longer a free-floating pretty number but a lattice invariant of the framework; P35 narrows from "where does the constant come from?" to "why does the exponent 36 couple to the packing density". The condition of witness B becomes considerably weaker thereby, but does not vanish.

Witness C: the power form (A040)

[K] The power form $(D_f^{\text{eff}}/3)^{D_f^{\text{eff}}/2}$ with $D_f^{\text{eff}} = 2.973$ yields 0.986651: residual additive $1.6 \cdot 10^{-5}$, multiplicative $1.0 \cdot 10^{-4}$ – tends additive. However, the four-digit rounding of D_f^{eff} ($\pm 2.5 \cdot 10^{-4}$ on K) exceeds the form distance: not an independent witness, but consistent with B.

.6 Rolled up and unrolled: why the additive form has a reason

Section .5 treats the form question as a measurement problem. This section treats it as a *structural problem*: the two forms are not two arbitrary computational ansätze but the exact bookkeepings of *two different representations of the same geometry*. The form question thereby becomes a precise physical question: In which domain is the correction exactly defined – rolled up or unrolled?

The two domains

[B] *Rolled up* (T^4 , winding picture). On the compact cycle the natural quantity is the phase or arc length, and phases *add*. A correction that shortens each winding by the segment ε sums linearly over N windings. The closure condition from A040,

$$75(1 - \varepsilon) = 74 \quad \Rightarrow \quad \varepsilon = \frac{1}{75} = 100\xi,$$

is *exact* in this domain – as a bookkeeping identity of the winding, not as an approximation: 75 shortened revolutions close precisely when one full revolution is missing in total. The additive form $K = 1 - 100\xi$ is therefore not a linearisation of anything; it is the native, closed bookkeeping of the rolled-up geometry – representable in exact fraction arithmetic because ξ is rational.

[B] *Unrolled* (scale flow, measurement domain). Unrolling the cycle in the scale direction turns the winding count into a run over orders of magnitude (the RG flow from A040, ~ 17 decades). In this representation corrections act stepwise on amplitudes, and stepwise amplitude corrections compose multiplicatively: $(1 - \xi)^{100} \approx e^{-100\xi}$. Were the unrolled description fundamental, *this* form would be exact and the additive only its first order.

The form distance is the translation error

[B] By Section .4 the entire form distance is the second-order binomial term, $4950\xi^2$. It is thus nothing other than the error of the translation between the domains at second order. The form question reads sharply: Which domain carries the exact definition, which the approximation? Precisely this – and only this – is what the witnesses measure; currently only witness B resolves it.

The Euler parallel makes the assignment unambiguous

[B] The musical spiral provides the control case in which the domain assignment is *uncontested* – because there, measurement takes place. What ear and instrument reach are frequencies, i.e. quantities of the unrolled scale domain; intervals compose multiplicatively there ($3/2 \cdot 3/2 \cdot \dots$), and only the logarithm rolls them onto the octave circle – whereby the irrationality enters and prevents closure in principle. The ξ cycle, conversely, is *defined* in the rolled-up domain: from the outset a circle structure with rational winding number, without two multiplicatively defined objects that would first have to be made comparable logarithmically.

	Music (Euler)	FFGFT (ξ cycle)
Measured quantity	frequency (unrolled)	— (defined rolled up)
Composition	intervals multiply	winding phases add
Bridge to the circle	logarithm (forced)	none needed
Step size	$\log_2(3/2)$ irrational	$1/75$ rational
Closure	never (commas only)	exact after 75

The assignment is complete: multiplicative composition belongs to the unrolled measurement/scale domain, additive composition to the rolled-up winding domain.

The corpus already contains the fork: case B against case C

[K] The assignment does not stand beside the corpus but within it. The closure fork (Docs. 295/313, cited in Doc. 314) carries three cases: case A ($d = 0$, closure after one revolution), case B (ξ frozen: rational rotation number $1 - d = 74/75$; because $\gcd(74, 75) = 1$, closure after exactly 75 revolutions) and case C (ξ runs via the recursion: “the cumulated defect becomes logarithmic and the winding an equiangular spiral” – no closure). This is precisely the form question as physics: case B is the additive bookkeeping (linear defect, rational winding, exact closure), case C is the multiplicative–logarithmic composition – and the equiangular spiral is the same mathematical picture as the Euler spiral of control case I. The two forms of K_{frak} are thus the bookkeepings of the two branches of the fork; the question “additive or multiplicative?” is the question of which branch carries the correction.

Doc. 314 (Ch. D2) additionally sharpens the domain boundary in the terminology used here: *in the rolled-up state there are no fractal defects; the defect $d = 100\xi$ accumulates only unrolled*, and K_{frak} is a property of the traversal, entering via the bridge constants (E_0, ν), not via the local operator. The additive form is accordingly the bookkeeping of the closed traversal with frozen ξ (case B); the multiplicative would be the bookkeeping of the running accumulation (case C). All corpus uses of K_{frak} (A040 rule, R70 assignment, R72) book case B.

What the witnesses then mean

[B] In this reading, witness B does not measure a computational detail but the domain membership of the correction – and it answers: the exact bookkeeping is the *rolled-up* one; witness C tends the same way. The multiplicative form is the approximation a naive observer of the unrolled scale narrative would write down – correct to first order, wrong from $4950\xi^2$ on.

This also resolves the apparent tension within A040: the document contains both – the closure comma (rolled-up, exact reading) and the cumulative RG flow (unrolled derivation of the factor 100). Both are consistent because they agree to first order; by the witness evidence, the rolled-up one is exact. The RG flow derives *why* the correction has magnitude 100ξ ; the winding geometry fixes *how* it composes. The naive RG argument for the multiplicative form is thereby placed rather than refuted: it is the correct argument of the unrolled domain – and that domain, by the witness evidence, is not the defining one.

Connection to the foundational architecture

[B] The assignment “rolled up = exact” is not an ad-hoc rescue of the additive form but congruent with the foundational architecture of the framework: the compact geometry (T^4 , rational ξ stipulation) is primary; scale-flow narratives are derived descriptions. That

the exact arithmetic of the corpus works throughout with fractions is the same statement in methodological form: exact fractions exist only in the rolled-up domain – the unrolled one knows only approximations with ξ^2 residuals. In this respect a consistently rolled-up framework *must* carry the additive form; were the measurement to come out multiplicative, not only K_{frak} would be affected but the primacy claim of the geometry itself.

.7 Status and falsification line

Bookkeeping status

$K_{\text{frak}} = 1 - 100\xi = 74/75$: value **[K]** (A040). Form additive: **[B]**, once conditionally confirmed – the condition ($16/\pi^2$ reference, witness B) is substantially weakened by Doc. 314, since $16/\pi^2 = 1/\Delta(\text{D4})$ is identified corpus-natively and only the coupling to the exponent 36 remains open (P35 narrowed). Structurally the additive form is the bookkeeping of fork branch case B (frozen ξ , rational winding), which all corpus uses book; the multiplicative alternative $(1 - \xi)^{100}$ is the bookkeeping of the spiral branch case C, is disfavoured by the conditional witness B and supported by no witness.

[B] Why nevertheless not [K]-measured: The only sharp discrimination (witness B) hangs on the status of $16/\pi^2$ as reference (P35, narrowed). If this premise later turns out correct, the form statement becomes unconditional. Conversely, the multiplicative form remains logically possible as long as it is open: then the $16/\pi^2$ proximity would be a coincidence – and the naive RG argument of the unrolled domain would have been right.

[B] Falsification line. The question would be decidable unconditionally at a location where K stands at high power against a *measurement* rather than a coincidence constant. The candidate already stands in the corpus: the baryon prediction from A270 (sector exponent +2, i.e. K^{38} level, form distance there $\approx 0.34\%$). A baryon mass check of better accuracy would decide additive against multiplicative without conditions – and at the same time test the domain assignment of Section .6: were it to come out multiplicative, not only the form choice would be corrected; the correction would live in the scale domain, against the winding reading of the closure comma. The structural argument thus raises the stakes; it binds the form question to the primacy question.

.8 Open items

- **P-315-2:** The residual of the α point: α requires $K_{\text{frak}} = 0.98650$, i.e. $1.7 \cdot 10^{-4}$ from $74/75$ (in n language 101.2 instead of 100). A correction term of the theory or a pole-mass effect? Consistent with R72.
- **P35 (narrowed by Doc. 314):** With $16/\pi^2 = 1/\Delta(\text{D4})$ the constant is identified as the reciprocal D4 packing density; what remains open is the forward derivation of the coupling $K_{\text{frak}}^{-36} = 1/\Delta(\text{D4})$, in particular the role of the bulk exponent 36.
- **P-315-3 (n sharpness):** In n language, form discrimination requires $\Delta n \leq 0.66$ ($= n(n-1)\xi/2$). The three n routes from R72 (101.2; 100 ± 2 ; 100.27) do not reach this; the sector ladder (100.27) is the most promising candidate for sharpening.
- **Sharpening A040:** state D_f^{eff} with more digits (needed: better than $\pm 1 \cdot 10^{-4}$) so that witness C becomes resolution-capable.
- **Baryon location (A270):** mass check at the K^{38} level as unconditional form test – simultaneously a test of the fork assignment case B/C.

.9 Verification

All numbers of this document are reproducible by the following scripts (pure standard library; exact fraction arithmetic where possible):

- `python/315_Skripte/euler_spirale_7limit.py` – comma search in the 5/7-limit, exact; assertion against exact closure.
- `python/315_Skripte/pruefrechnung_kfrak_form.py` – witnesses A/B/C with verdict logic (witness A via the K_{frak} required by α_i uncertainty against the effective, possibly amplified form distance).

Inputs: $\xi = 4/30000$ exact; $m_e = 0.51099895069$ MeV, $m_\mu = 105.6583755$ MeV (CODATA 2022).